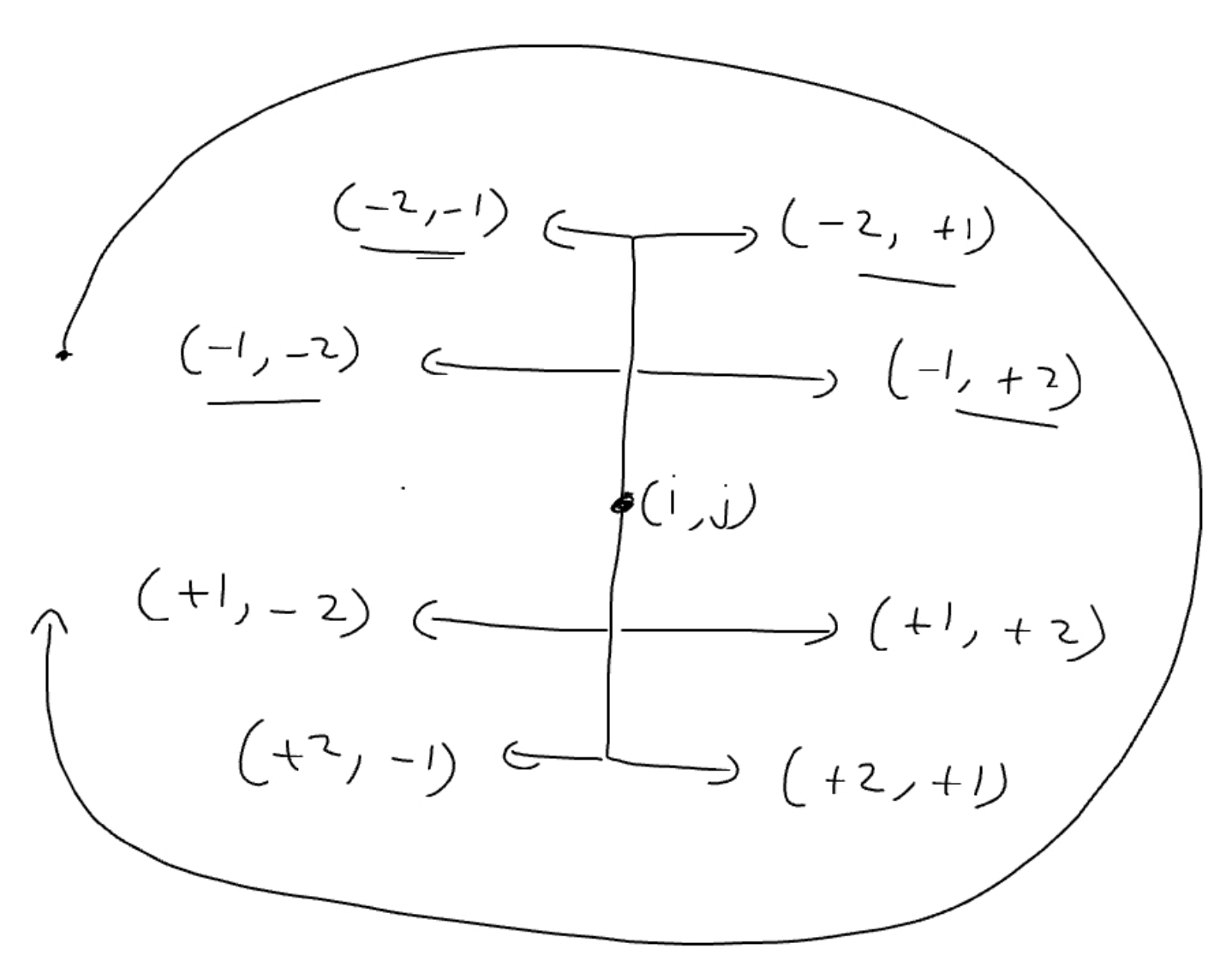
0 → 1, 2
1 →
2 →

0 → 2, 1
1 →
2 →

[0, 2, 1]

[0, 1, 2]


(-2, -1)
(-2, +1)
(-1, -2)
(-1, +2)
(i, j)
(+1, -2)
(+1, +2)
(+2, -1)
(+2, +1)

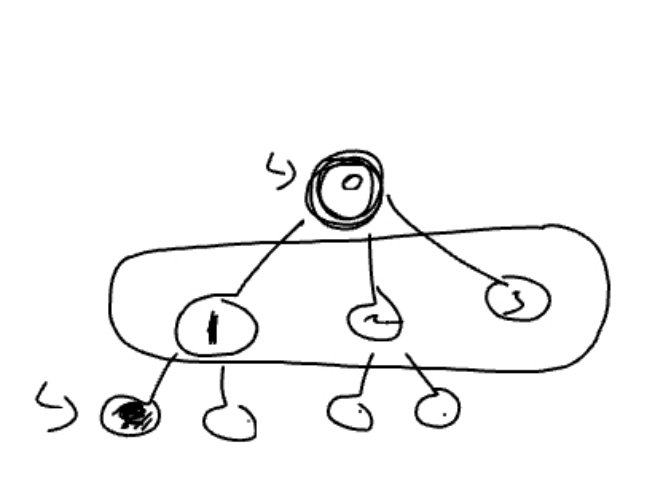
n × n

6×6.

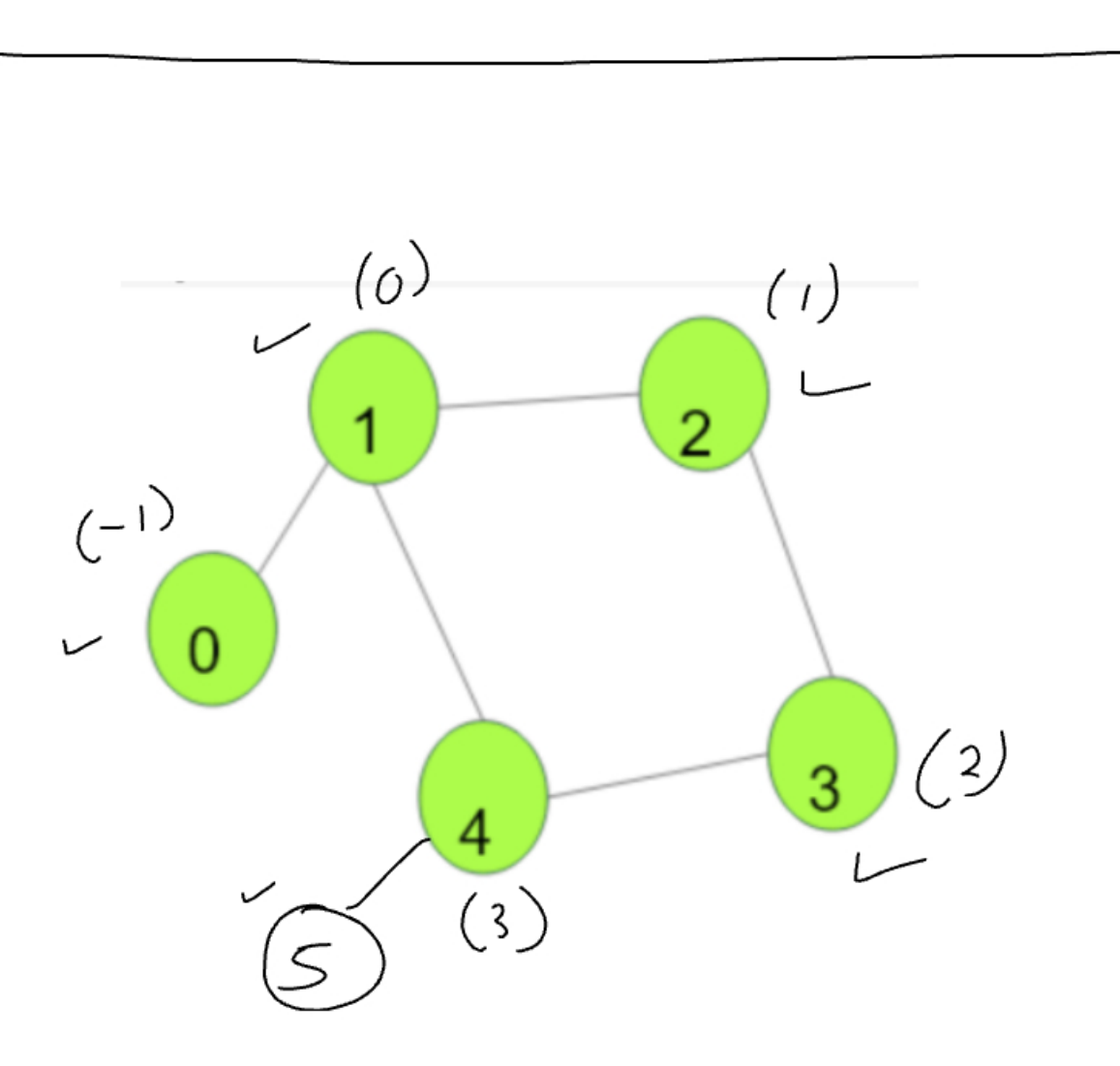
~~1, 2~~ ~~3~~, 2

0 1, 2, 3

res : 0 1 2 3



SSSP

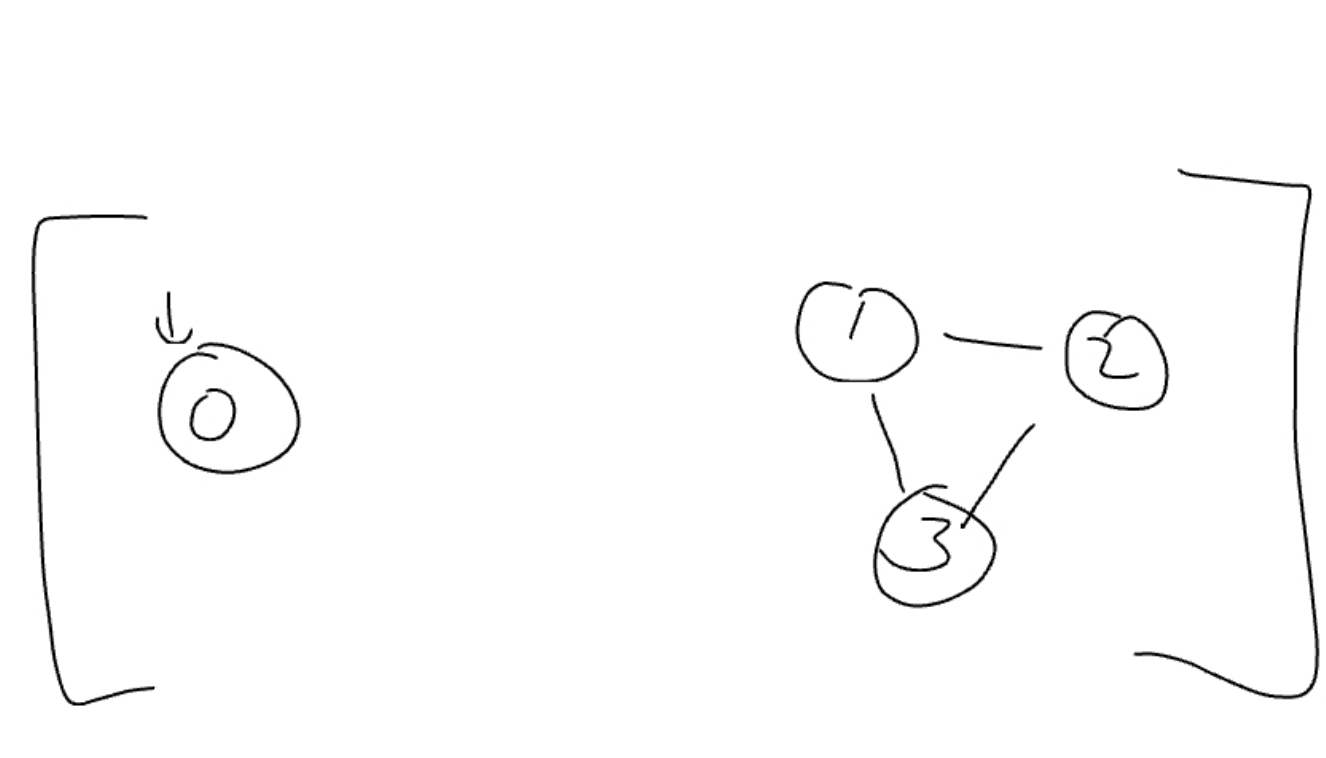

(0)
1
(1)
2
(-1)
0
3
(2)
4
(3)
5

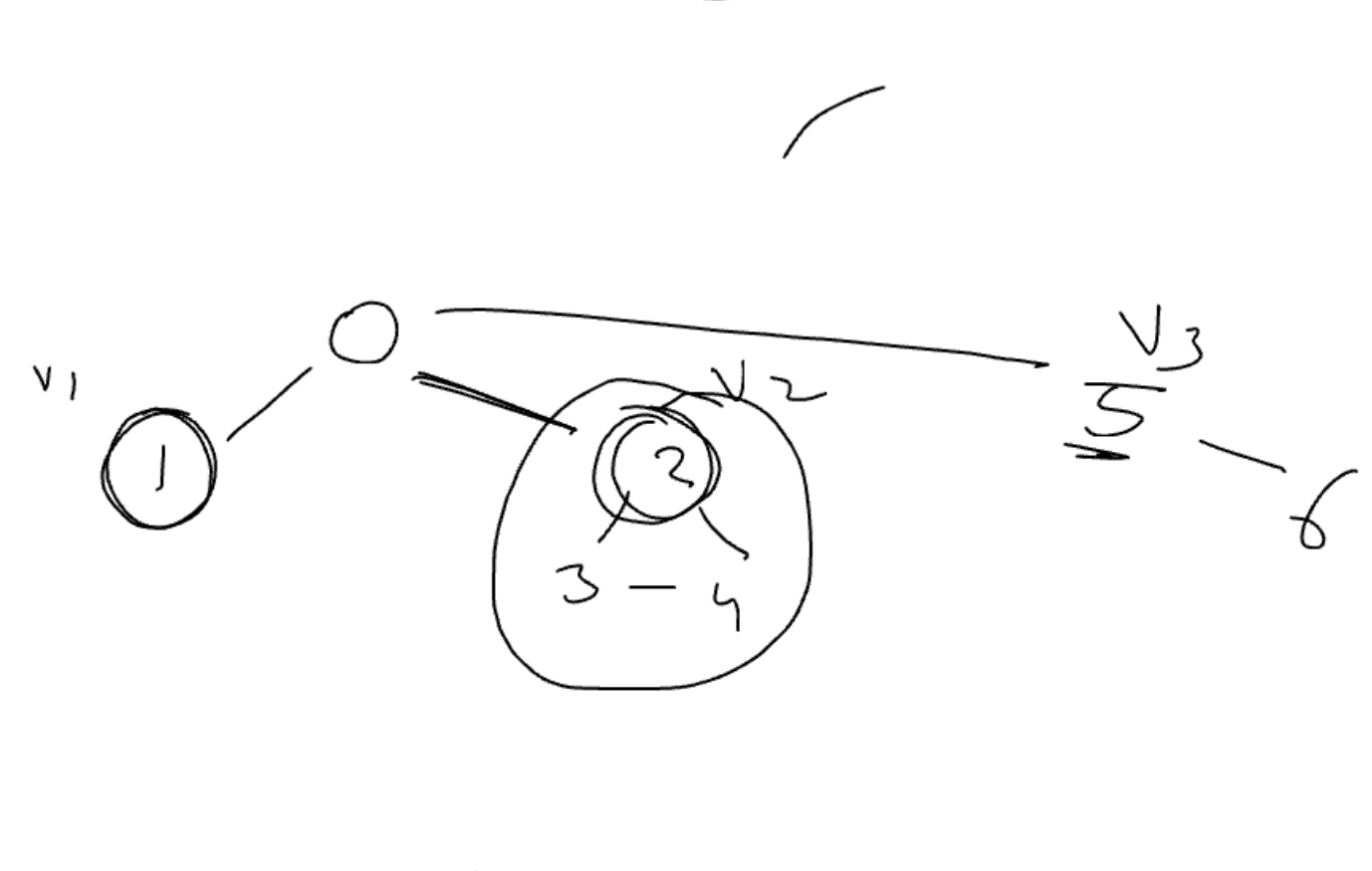
TC : O(V+E)
AS : O(V)

0 — 1 (parent - 0)

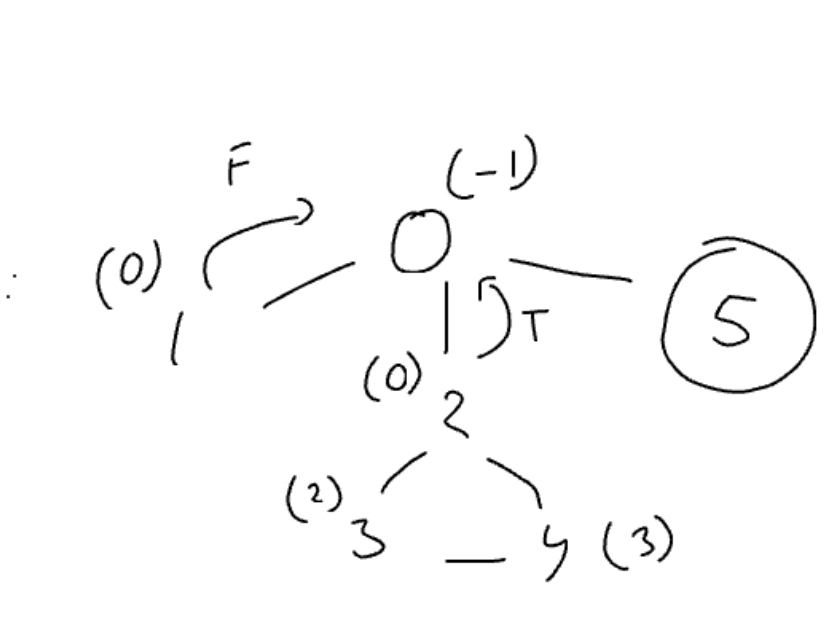
(-1) 0 → 1 (0)

0→0

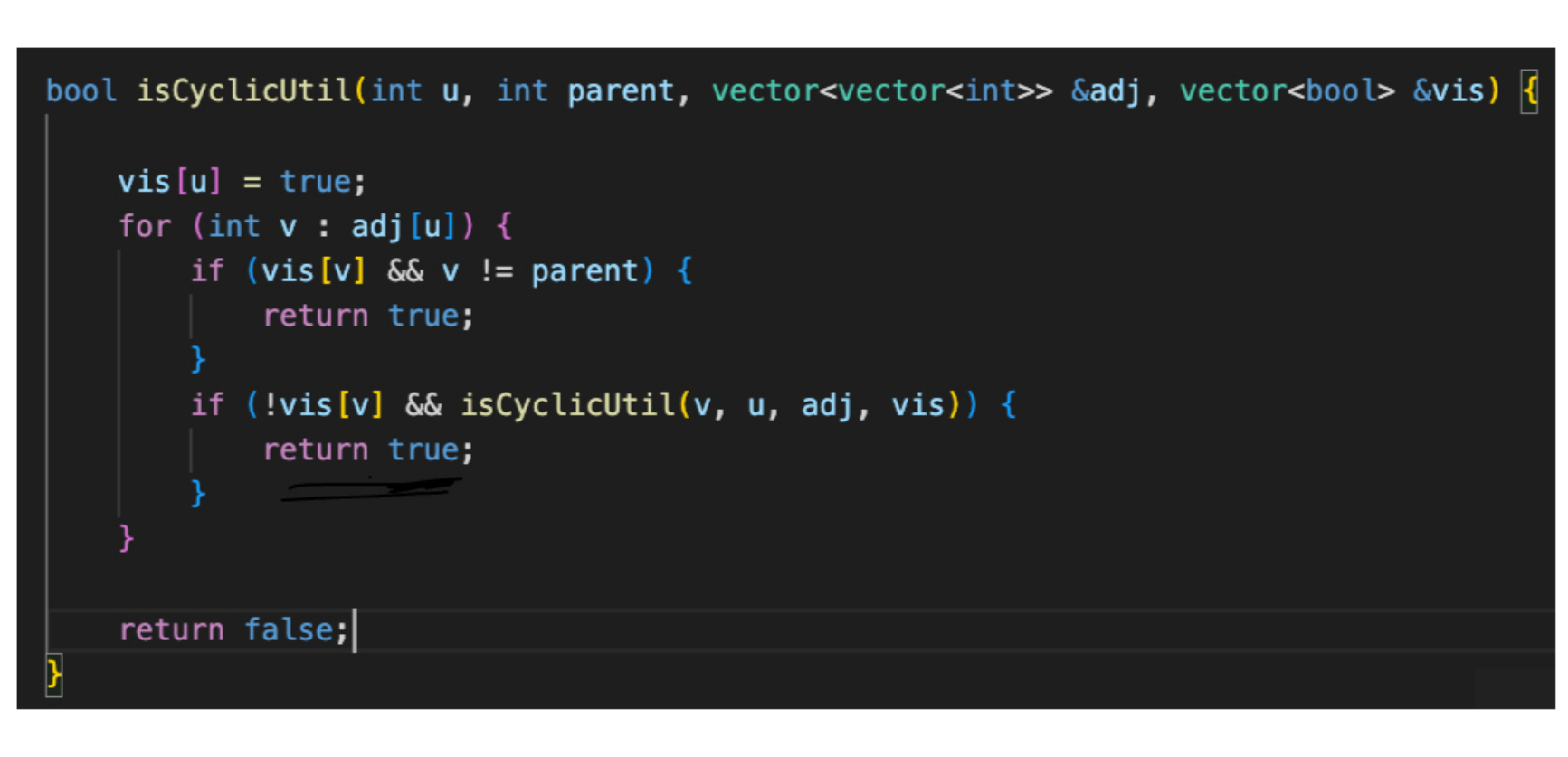



V1
1
0
2
V2
3
4
V3
5
6

u = 0 F
u = 1
u = 2 T
u = 3 T
u = 4


F
(-1)
(0)
1
0
T
5
(0)
2
(2)
3
4
(3)

0 → 1, 3, 5
1 → 0
2 → 0, 3, 4
3 → 2, 4
4 → 2, 3
5 → 0

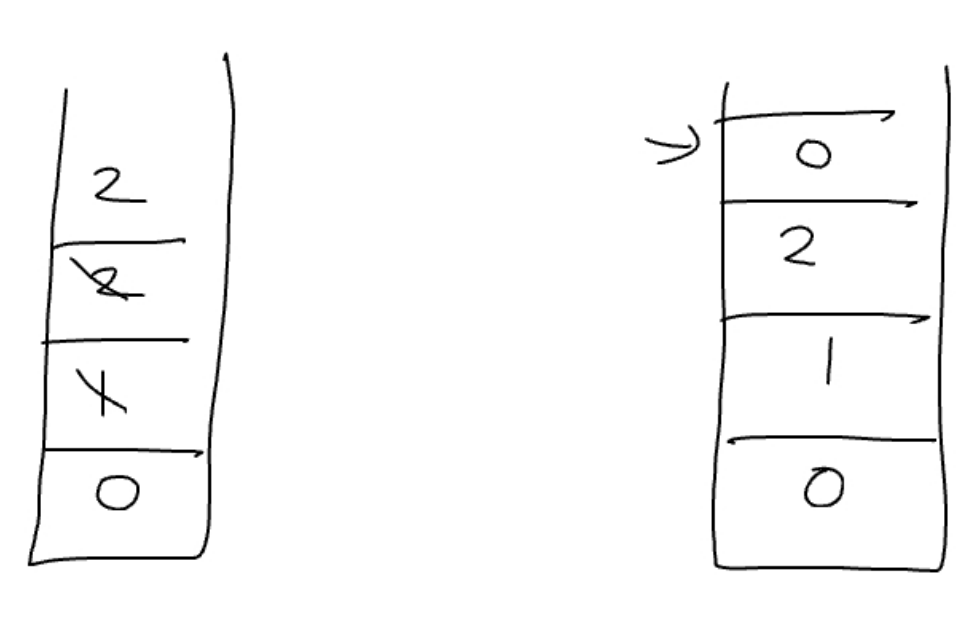

```
bool isCyclicUtil(int u, int parent, vector<vector<int>> &adj, vector<bool> &vis) {

    vis[u] = true;
    for (int v : adj[u]) {
        if (vis[v] && v != parent) {
            return true;
        }
        if (!vis[v] && isCyclicUtil(v, u, adj, vis)) {
            return true;
        }
    }

    return false;
}
```

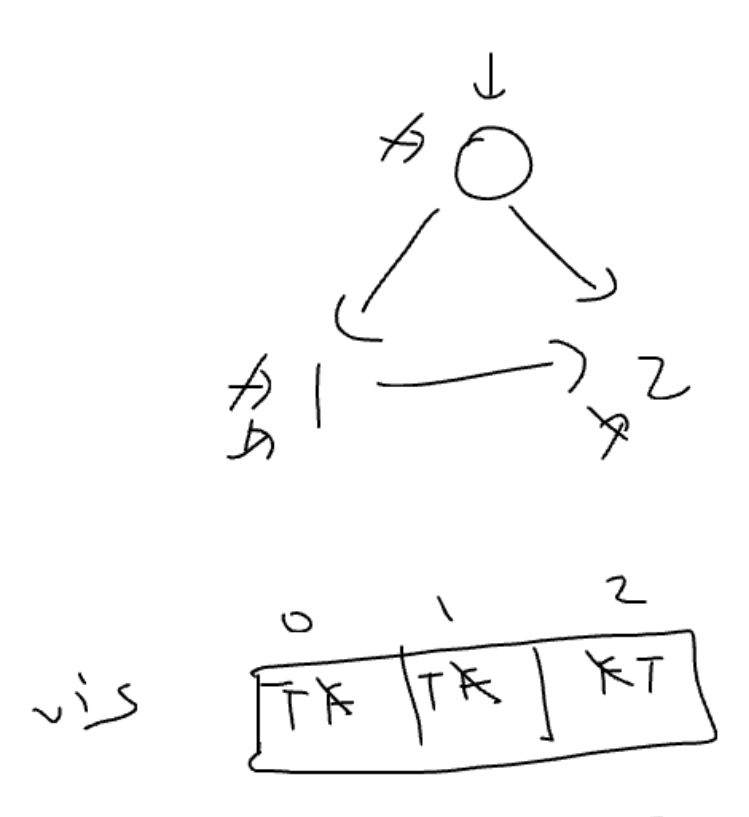
0

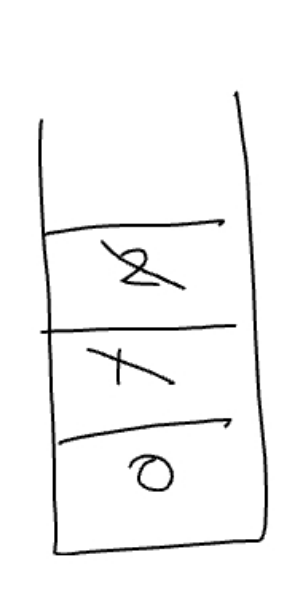
0 → 1, 2
1 → 2
2 →

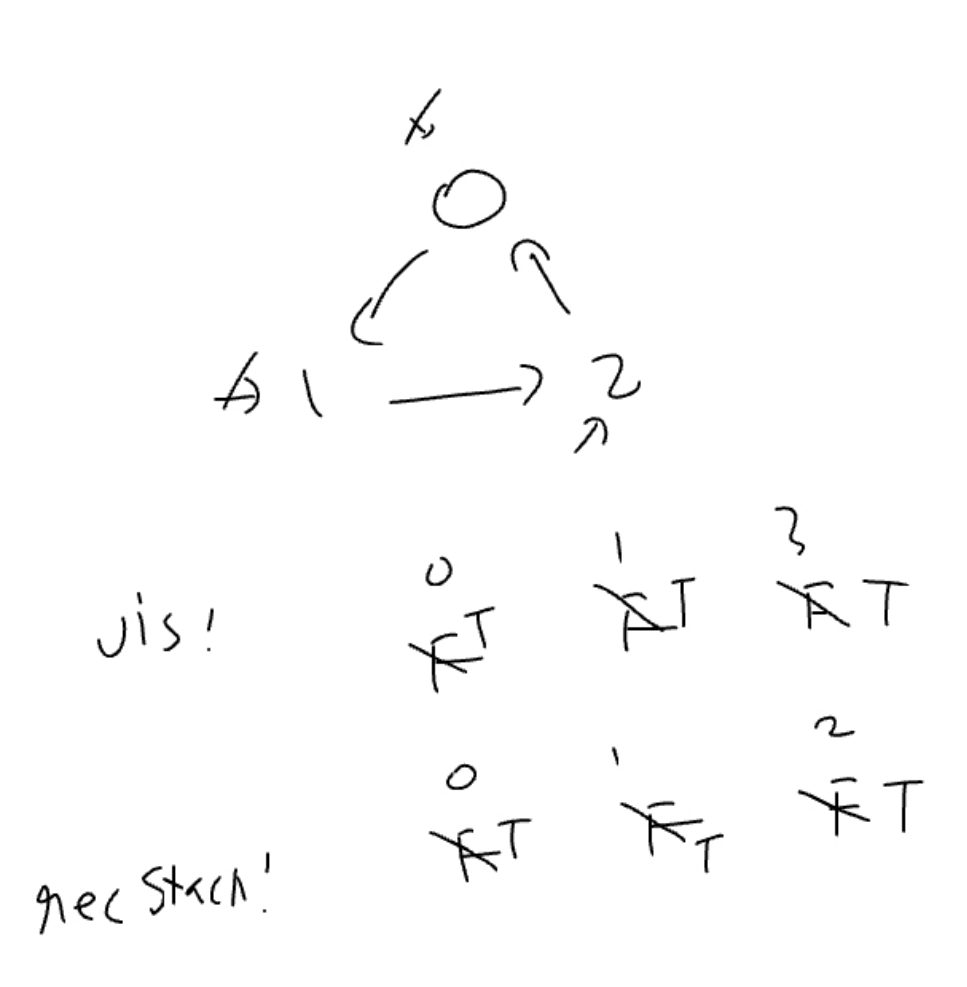


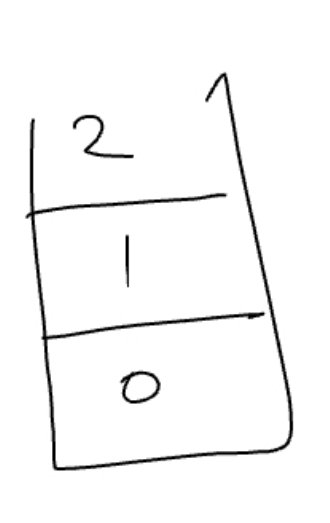
~~0, 1, 2,~~
~~0→1~~

[0→1


0
1
2
0 1 2
vis
~~F~~T ~~F~~T ~~F~~T
0 1 2
recStack
~~F~~T ~~F~~T ~~F~~F




0
1
2
vis:
0 ~~F~~T
1 ~~F~~T
2 ~~F~~T
recStack:
0 ~~F~~T
1 ~~F~~T
2 ~~F~~T

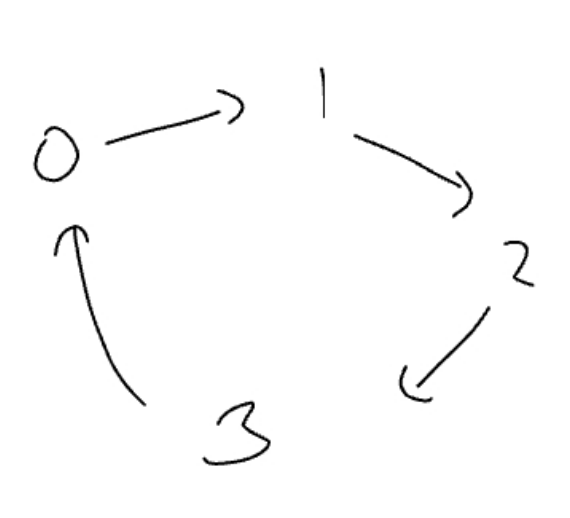


TC : O(V+E)
AS : O(V)

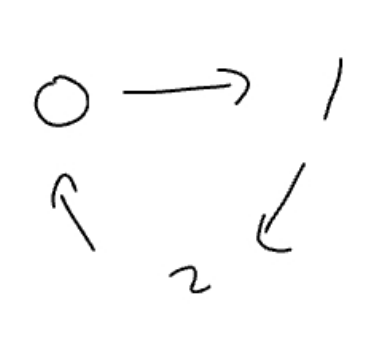
Topological Sort

u → v

u should come before v

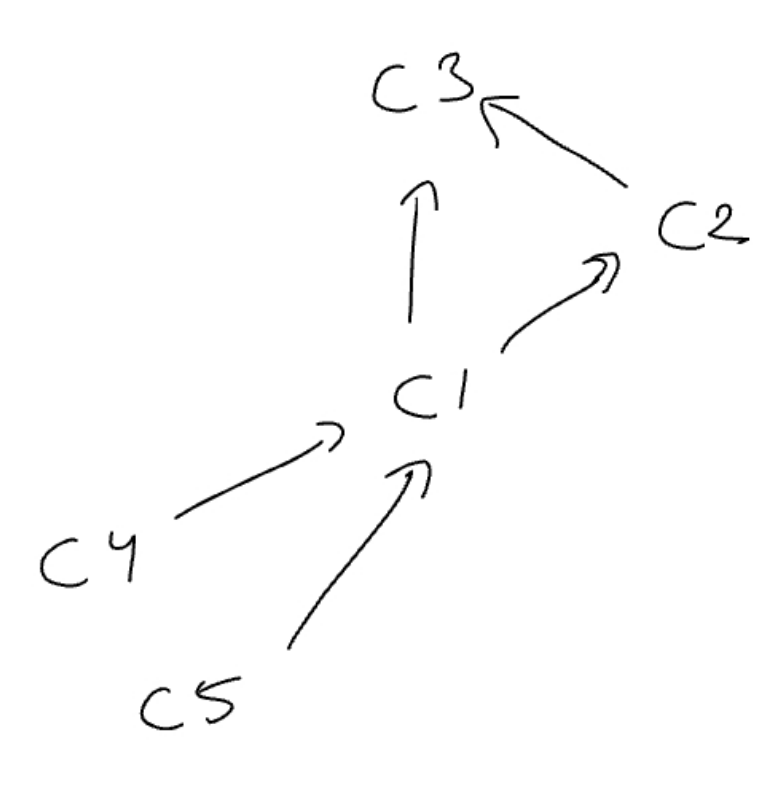


0, 1, 2, 3 X

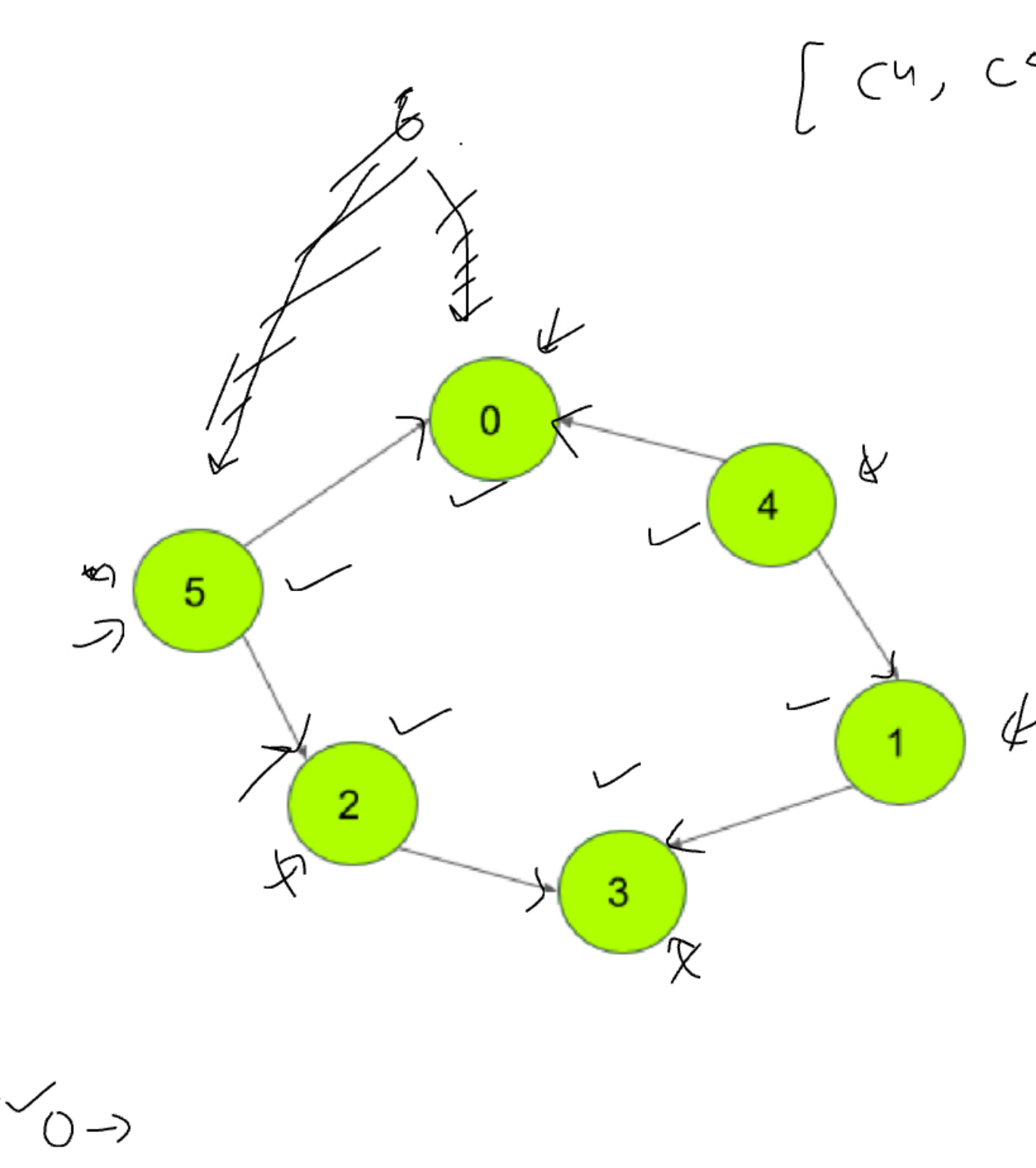


0, 1, 2 X

C1 → C4, C5
C2 → C1
C3 → C1, C2
C4
C5


C3
C2
C1
C4
C5

[C4, C5, C1, C2, C3]



X
U → v1, v2, v3 ...

[v1, v2, v3, U X
v1, v2, v3, U, X]

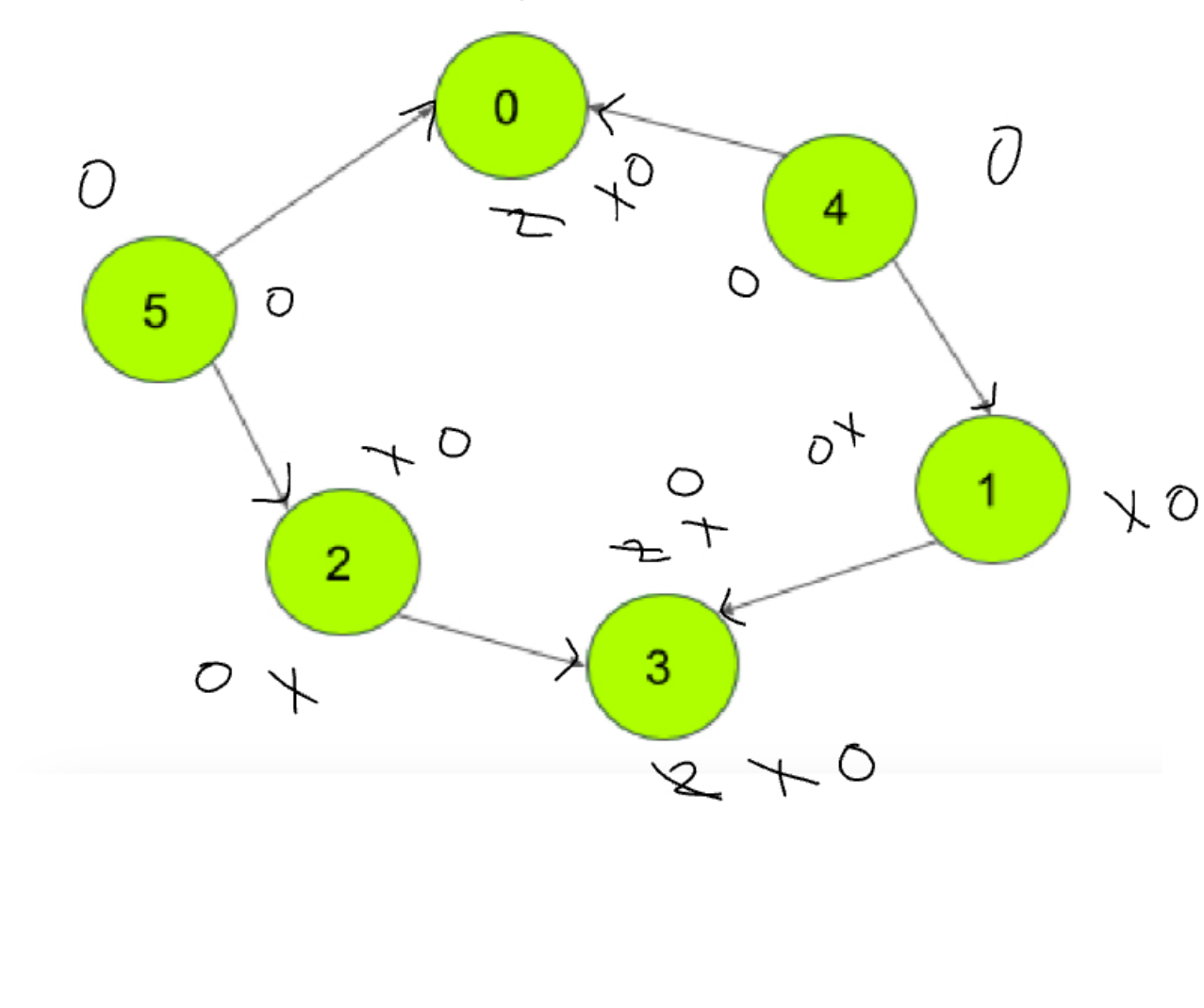
5

res: [0, 3, 1, 2, 4, 5]

res: [5, 4, 2, 1, 3, 0]

0 →
1 → 3
2 → 3
3 → —
4 → 0, 1
5 → 0, 2

TC : O(V+E)
AS : O(V)


~~2~~ X 0
0
~~2~~ X 0
0
4
0
5
0
0
X 0
0 X
1
X 0
~~2~~ X
2
0 X
3
~~2~~ X 0

0

res: [4, 5, 1, 0, 2, 3]

queue : ~~4~~, ~~5~~, ~~1~~, ~~0~~, ~~2~~, ~~3~~

~~4~~ 5

queue : [~~4~~, ~~5~~, ~~1~~, ~~0~~, ~~2~~, ~~3~~

[4, 5, 1, 0, 2, 3]

TC : O(V+E)
AS : O(V)